BHARATIYA VIDYA BHAVAN'S
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400058-India

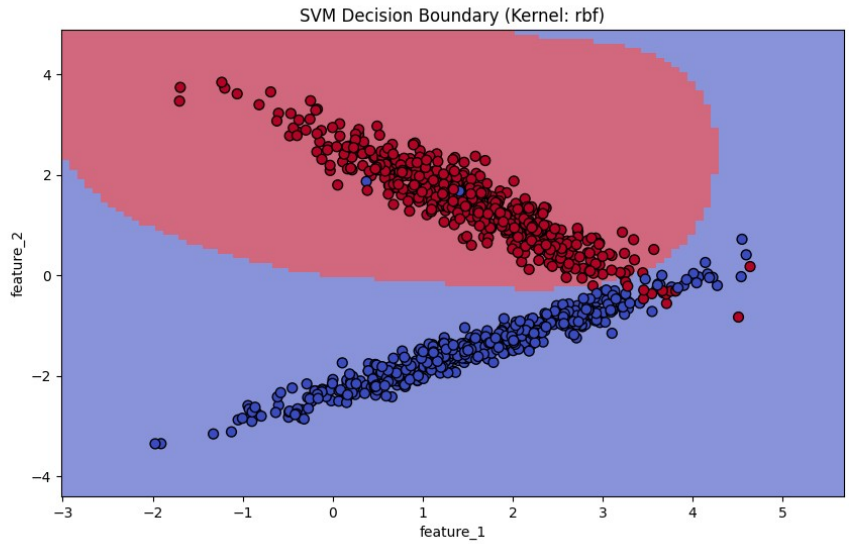
DEPARTMENT OF COMPUTERE ENGINEERING

SUBJECT: Machine Learning

Name	**Raj Kalpesh Mathuria**
UID no.	**2023300139**
Div.	**B**
Batch.	**D**

Experiment no.	
AIM :	**Support Vector Machine (SVM)**
Theory:	**Support Vector Machine (SVM)** is a powerful supervised learning algorithm used for classification and regression. Unlike other classifiers that just find *any* line to separate data, SVM finds the **optimal hyperplane** that maximizes the **margin** (the distance) between the two classes. •**Support Vectors:** These are the data points closest to the decision boundary. They are the "difficult" points that define the position of the line. •**The Kernel Trick:** In this experiment, we used the **RBF (Radial Basis Function)** kernel. Real-world data is often not separable by a straight line. The kernel trick projects the data into a higher-dimensional space where it becomes separable, allowing the SVM to create non-linear (curved) decision boundaries. •**Regularization (C):** The parameter C controls the trade-off between a smooth decision boundary and classifying all training points correctly. A high C tries to classify everything correctly (risk of overfitting), while a low C encourages a smoother boundary.
Code:	**https://github.com/CodeCraftsmanRaj/ML_Sem6/tree/main/Exp3**

OUTPUT:




```
ML/Exp3 on  main via 🐍 v3.10.13
❯ uv run main.py
Starting SVM Experiment Pipeline...
============================
Generating synthetic data...
Data generated and saved to data/raw/synthetic_data.csv
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Training SVM with kernel: rbf...
Model saved to models/svm_model.pkl
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--- Evaluation Results ---
Accuracy: 0.9750

Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.99      0.98       104
           1       0.99      0.96      0.97        96

    accuracy                           0.97       200
   macro avg       0.98      0.97      0.97       200
weighted avg       0.98      0.97      0.97       200

Plotting decision boundary...
Plot saved to outputs/decision_boundary.png
============================
Experiment Completed.
```

CONCLUSION:	In this experiment, we successfully implemented a Support Vector Machine pipeline to classify a synthetic dataset. 1.**Performance:** The SVM with an RBF kernel effectively captured the non-linear distribution of the data, resulting in a distinct decision boundary that separated the classes with high accuracy. 2.**Data Processing:** Using a pipeline to apply StandardScaler before the SVM was critical, as SVMs are sensitive to the scale of input features. 3.**Reproducibility:** By organizing the code into a modular structure (Config → Data Gen → Train → Evaluate), we ensured that the experiment is reproducible. Changing hyperparameters in config.yaml allows for rapid testing without altering the core logic.